Course Notes

for (CSED490V) Foundations of Continuous Computation

Jungwoo Kim
Department of Mathematics
Pohang University of Science and Technology

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Notation and Conventions

General notation and writing follow *(MATH000) Mathematical Language*.

Notation	Meaning
$\mathbb{R}_C$	Computable real numbers
P_{root}	Integer polynomials admitting an integer zero
$\mathcal{A}$	Oracle algorithm
$\gamma_{E,M}$	Floating-point representation map
$\mathbb{F}_{E,M}$	Finite floating-point values
$\overline{\mathbb{F}}_{E,M}$	Extended floating-point values
$e_{\min},\ e_{\max}$	Minimum and maximum normal exponents
$\max \mathbb{F}_{E,M},\ F_{\max}$	Largest finite floating-point value
$K_{E,M}$	Overflow threshold (rounding to infinity)
$\text{rnd}_{E,M}$	Round to nearest, ties to even
$\oplus,\ \ominus$	Rounded addition and subtraction
$\otimes,\ \odot$	Rounded multiplication
$\oslash,\ \oslash$	Rounded division
fma	Fused multiply-add (one rounding)
$\text{err}_{\text{abs}},\ \text{err}_{\text{rel}}$	Absolute and relative errors
$\text{ulp},\ \text{err}_{\text{ulp}}$	Unit in the last place and ULP error
ε	Unit roundoff, 2^{-M-1}
e_a	Exponent assigned to $a \in \mathbb{F} \setminus \{0\}$
fast2sum	Error-free summation algorithm
$\sigma,\ \Sigma$	Input valuation and admissible input set
$[\![e]\!],\ [\![e]\!]_{\text{r}}$	Floating-point and real-valued semantics
$\text{range}(e),\ \text{err}(e)$	Range enclosure and absolute-error bound over Σ

1 Computable Reals

Definition 1.1. Borel Computability
A real number r is *Borel-computable* if there exists a computable sequence $(q_n)_{n\in\mathbb{N}_0} \subseteq \mathbb{Q}$ such that

$$|r - q_n| < 2^{-n} \qquad (n \in \mathbb{N}_0).$$

Definition 1.2. Turing Computability
A nonnegative real number r is *Turing-computable* if it has a binary expansion

$$r = a + \sum_{i=0}^{\infty} b_i 2^{-(i+1)}, \qquad a \in \mathbb{N}_0, \quad b_i \in \{0,1\},$$

for which $i \mapsto b_i$ is computable. A negative real number is Turing-computable if its absolute value is Turing-computable. Dyadic rationals admit both terminating and repeating-one binary expansions; the terminating expansion may be used.

Theorem 1.3. Equivalence of Computability Definitions
For $r \in \mathbb{R}$, the following are equivalent:

(i) r is Borel-computable.

(ii) r is Turing-computable.

Proof. It suffices to consider $r \geq 0$, as changing the sign of rational approximations is effective and binary computability for negative numbers is defined through absolute values.

Suppose r has a computable binary expansion with integer part a and digits b_i. The computable rationals

$$q_n = a + \sum_{i=0}^{n} b_i 2^{-(i+1)}$$

satisfy

$$0 \leq r - q_n \leq \sum_{i=n+1}^{\infty} 2^{-(i+1)} = 2^{-(n+1)} < 2^{-n}.$$

Thus Turing computability implies Borel computability.

Conversely, let $|q_m - r| < 2^{-m}$ for a computable rational sequence. The integer part $a = \lfloor r \rfloor$ can be fixed in the existential specification of a binary expansion; this is not asserted to be a uniformly computable operation on rational names.

If r is dyadic, its terminating expansion has only finitely many nonzero digits and is therefore computable. Otherwise, for each $n \in \mathbb{N}_0$, the number r lies in the interior of a unique cell of mesh $2^{-(n+1)}$. Search for m and $k \in \mathbb{N}_0$ such that

$$(q_m - 2^{-m},\ q_m + 2^{-m}) \subset \left(\frac{k}{2^{n+1}}, \frac{k+1}{2^{n+1}} \right).$$

The containment can be checked using rational arithmetic. The search terminates because r is not a cell boundary and the intervals shrink to r. Since r lies in the returned interval, $k = \lfloor 2^{n+1} r \rfloor$; hence the nth binary digit is

$$b_n = k \bmod 2.$$

This computes the binary expansion in the non-dyadic case. □

Remark. These definitions determine the same real numbers, not uniformly interconvertible representations. A computable binary expansion yields rational approximations effectively, whereas no single algorithm can convert every rational name into binary digits: it would have to resolve dyadic boundary cases. This does not contradict the existence of some computable expansion for each Borel-computable real.

Definition 1.4. Computable Real Numbers

$$\mathbb{R}_C := \{r \in \mathbb{R} \mid r \text{ is computable}\}.$$

The preceding theorem allows either notion of computability.

Proposition 1.5. Closure under Arithmetic
The set $\mathbb{R}_C$ is a subfield of $\mathbb{R}$. In particular,

$$x, y \in \mathbb{R}_C \implies x + y,\ x - y,\ xy \in \mathbb{R}_C, \qquad y \neq 0 \implies \frac{x}{y} \in \mathbb{R}_C.$$

Proposition 1.6. Real Closedness of Computable Reals
Every real root of a nonzero polynomial in $\mathbb{R}_C[X]$ is computable. Consequently, $\mathbb{R}_C$ is real closed (not algebraically closed).

Proposition 1.7. Computability of π and e
The real numbers π and e are computable.

Proof. Using the alternating arctangent series,

$$\frac{\pi}{4} = \arctan 1 = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}.$$

For $m \in \mathbb{N}_0$ let

$$p_m = 4 \sum_{k=0}^{m} \frac{(-1)^k}{2k+1} \in \mathbb{Q}.$$

The alternating-series estimate gives $|\pi - p_m| \leq 4/(2m+3)$. Taking $m = 2^{n+1}$ yields $|\pi - p_m| < 2^{-n}$, and the finite rational sum is computable. The computability of $e = \exp(1)$ follows from the computability of $\exp\colon \mathbb{R} \to \mathbb{R}$ proved in Theorem 2.8. □

Proposition 1.8. Countability of the Computable Reals
The set $\mathbb{R}_C$ is countably infinite. Consequently, noncomputable real numbers exist.

Proof. Algorithms have countably many finite descriptions, so $\mathbb{R}_C$ is at most countable. Every rational is computable by a constant approximation sequence; hence $\mathbb{Q} \subseteq \mathbb{R}_C$ and $\mathbb{R}_C$ is infinite. Finally, $\mathbb{R}$ is uncountable. □

Fact 1.9. Hilbert's Tenth Problem
No algorithm decides whether an arbitrary multivariate integer-coefficient polynomial admits an integer zero. In particular, the collection of such polynomials having integer zeros is undecidable.

Proposition 1.10. A Noncomputable Real
Let P be the set of integer-coefficient polynomials in finitely many variables and let $\phi\colon P \to \mathbb{N}$ be a computable injection. Write $P_{\text{root}} \subseteq P$ for the polynomials admitting an integer zero. Then

$$r \coloneqq \sum_{p \in P} 2^{-\phi(p)} \mathbb{1}_{P_{\text{root}}}(p)$$

is a noncomputable real in $[0, 1]$.

Proof. Because ϕ is injective, each $2^{-\phi(p)}$ specifies a distinct binary position. Both P_{root} and its complement in P are infinite, so this expansion has infinitely many zeros and ones; r is not dyadic and its binary expansion is unique. If r were computable, its unique binary expansion would be computable by the equivalence theorem. Then

$$p \longmapsto b_{\phi(p)-1} = \mathbb{1}_{P_{\text{root}}}(p)$$

would decide Hilbert's Tenth Problem, a contradiction. □

Remark. The construction does not depend on P_{root}. For any noncomputable set $A \subseteq \mathbb{N}_0$, the real number

$$r_A \coloneqq \sum_{n=0}^{\infty} 2^{-(n+1)} \mathbb{1}_A(n)$$

is noncomputable. Indeed, A is neither finite nor cofinite, so the displayed expansion is unique; its computability would decide A.

2 Computable Functions

Definition 2.1. Computable Function
Let $D \subseteq \mathbb{R}$. A function $f\colon D \to \mathbb{R}$ is *computable* if an oracle algorithm $\mathcal{A}$ exists such that, for every $x \in D$, every rational name $\alpha\colon \mathbb{N}_0 \to \mathbb{Q}$ satisfying

$$|\alpha(i) - x| < 2^{-i} \qquad (i \in \mathbb{N}_0),$$

and every $n \in \mathbb{N}_0$, the algorithm with access to α and input n halts and returns $\mathcal{A}(\alpha, n) \in \mathbb{Q}$ with

$$|\mathcal{A}(\alpha, n) - f(x)| < 2^{-n}.$$

Only finitely many oracle queries are made in each computation. The definition extends componentwise to maps between subsets of $\mathbb{R}^k$ and $\mathbb{R}^m$.

Remark. The name α need not itself be computable. The same oracle algorithm must work for *every* valid rational name, including names of noncomputable reals. Unlike a binary-digit oracle, a rational name need not settle dyadic boundary cases.

If $f\colon D \to \mathbb{R}$ is computable, then $f(D \cap \mathbb{R}_C) \subseteq \mathbb{R}_C$: a computable name for $x \in D \cap \mathbb{R}_C$ can be supplied to the oracle for f.

Example 2.2. Basic Computable Functions
For $r \in \mathbb{R}_C$, the maps $x \mapsto x + r$ and $x \mapsto rx$ are computable. So are $|x|$, $\exp x$, $\sin x$, $\sqrt{x}$ on $[0, \infty)$, and $\log x$ on $(0, \infty)$. In several variables, $+$, $-$, $\times$, min, and max are computable, as is division on $\{(x, y) \in \mathbb{R}^2 \mid y \neq 0\}$.

Theorem 2.3. Closure under Composition
Let $f\colon D \to E$ and $g\colon E \to \mathbb{R}^m$ be computable, where $D \subseteq \mathbb{R}^k$ and $E \subseteq \mathbb{R}^\ell$. Then $g \circ f$ is computable.

Corollary 2.4. Computable Function Operations
If $f, g\colon D \to \mathbb{R}$ are computable, then so are

$$|f|, \quad f + g, \quad fg, \quad \min(f, g), \quad \max(f, g), \quad \exp \circ f.$$

Moreover, f/g is computable on $\{x \in D \mid g(x) \neq 0\}$.

Theorem 2.5. Computability Implies Continuity
Every computable $f\colon D \to \mathbb{R}^m$, with $D \subseteq \mathbb{R}^k$, is continuous on D in the relative topology.

Corollary 2.6. Discontinuous Tests
The sign function $\operatorname{sgn}\colon \mathbb{R} \to \{-1, 0, 1\}$ *and the predicates $x < y$ and $x = y$, interpreted as $\{0, 1\}$-valued functions on $\mathbb{R}^2$, are not computable on their full domains.*

Remark. Continuity is necessary but not sufficient for computability: for $c \in \mathbb{R} \setminus \mathbb{R}_C$, the constant function $x \mapsto c$ is continuous but not computable. A discontinuity at a point in the domain therefore prevents computability there.

Example 2.7. Discontinuous Functions and Their Domains
The step function $x \mapsto \mathbb{1}_{\{x>0\}}$ is not computable on $\mathbb{R}$, but is computable on $\mathbb{R} \setminus \{0\}$: refine the rational approximation until its sign is certified. The same principle makes strict comparison computable on $x \neq y$.

Alternatively, outside the single-valued framework above, one may allow a set of admissible outputs at the discontinuity:

$$\operatorname{sgn}'(x) = \begin{cases} \{1\}, & x > 0, \\ \{0\}, & x < 0, \\ \{0, 1\}, & x = 0. \end{cases}$$

This is a multivalued specification, not a discontinuous single-valued function made computable on its original domain.

Theorem 2.8. Computability of the Exponential Function
The function $\exp\colon \mathbb{R} \to \mathbb{R}$ *is computable.*

Proof. First let $x \in [-\frac{1}{2}, \frac{1}{2}]$ and let α be a rational name. For prescribed $n \in \mathbb{N}_0$, choose $m = d = n + 3$ and put

$$\widehat{x} = \alpha(m), \qquad P_d(t) = \sum_{j=0}^{d} \frac{t^j}{j!}, \qquad \widehat{y} = P_d(\widehat{x}).$$

This is a finite rational computation. Since $|x - \widehat{x}| < 2^{-m}$ and $|\widehat{x}| < 1$, the mean value theorem and $e < 3$ imply

$$|e^x - e^{\widehat{x}}| < 3 \cdot 2^{-m}.$$

For $d \geq 3$, Taylor's remainder is bounded by

$$|e^{\widehat{x}} - P_d(\widehat{x})| \leq \sum_{j=d+1}^{\infty} \frac{1}{j!} \leq \sum_{j=d+1}^{\infty} 2^{-j} = 2^{-d}.$$

Consequently,

$$|e^x - \widehat{y}| < 4 \cdot 2^{-(n+3)} < 2^{-n}.$$

For an arbitrary $x \in \mathbb{R}$, query $\alpha(0)$ and choose $k \in \mathbb{N}_0$ with $(|\alpha(0)| + 1)/2^k \leq \frac{1}{2}$. Then $|x/2^k| < \frac{1}{2}$, and a rational name of $x/2^k$ is obtained by scaling the oracle answers. Write $u = e^{x/2^k} \in (0, 2)$ and $N = 2^k$. For any rational v with $|u - v| < 1$,

$$|u^N - v^N| \leq N 3^{N-1} |u - v|.$$

Requesting a sufficiently accurate v and computing the rational v^N therefore approximates e^x to arbitrary precision. □

Theorem 2.9. Noncomputability of Strict Comparison
The map

$$(x, y) \longmapsto \begin{cases} 1, & x < y, \\ 0, & x \geq y \end{cases} \qquad (x, y \in \mathbb{R})$$

is not computable.

Proof. If the map were computable, requesting error less than $1/4$ and comparing the rational answer with $1/2$ would decide its Boolean output exactly. Consider the rational names

$$\alpha(i) = 0, \qquad \beta(i) = 2^{-(i+1)} \qquad (i \in \mathbb{N}_0),$$

both naming 0. The decision algorithm returns 0 after only finitely many oracle queries. Let N be at least every index queried from β, and define

$$\beta'(i) = \begin{cases} 2^{-(i+1)}, & i \leq N, \\ 2^{-(N+1)}, & i > N. \end{cases}$$

It is a valid name for $y' = 2^{-(N+1)} > 0$, because $|\beta'(i) - y'| < 2^{-i}$ for $i \leq N$ and the error is zero otherwise. The finite computation sees exactly the same answers on (α, β') and (α, β), so it still returns 0, whereas $0 < y'$ requires output 1. □

Theorem 2.10. Comparison of Unequal Real Numbers
Strict comparison is computable on

$$\{(x, y) \in \mathbb{R}^2 \mid x \neq y\}.$$

Remark. When $x \neq y$, sufficiently accurate rational approximations must eventually certify their order. Without this a priori promise, neither equality nor strict comparison is decidable on all of $\mathbb{R}^2$: their indicator maps are discontinuous along $x = y$.

3 Floating-Point Arithmetic

Definition 3.1. Floating-Point Representation
Fix $E \geq 2$ and $M \geq 1$. A floating-point bit string has a sign bit s, E exponent bits, and M fraction bits. Write

$$k = (e_1 \cdots e_E)_2, \qquad t = (0.m_1 \cdots m_M)_2 = \sum_{j=1}^{M} m_j 2^{-j}, \qquad b_E = 2^{E-1} - 1.$$

The representation map $\gamma_{E,M} \colon \{0,1\}^{1+E+M} \to \mathbb{R} \cup \{-\infty, +\infty, \mathrm{NaN}\}$ is defined by

$$\gamma_{E,M}(s, e_1, \ldots, e_E, m_1, \ldots, m_M) = \begin{cases} (-1)^s 2^{k-b_E}(1+t), & 1 \leq k \leq 2^E - 2, \\ (-1)^s 2^{1-b_E} t, & k = 0, \\ (-1)^s \infty, & k = 2^E - 1,\ t = 0, \\ \mathrm{NaN}, & k = 2^E - 1,\ t \neq 0. \end{cases}$$

The first case gives *normal* values; the second gives *subnormal* values when $t \neq 0$, and zero when $t = 0$. The remaining cases represent infinities and NaN. The normal exponent bounds are

$$e_{\min} = 1 - b_E = 2 - 2^{E-1}, \qquad e_{\max} = 2^E - 2 - b_E = b_E.$$

Define

$$\overline{\mathbb{F}}_{E,M} = \operatorname{im} \gamma_{E,M}, \qquad \mathbb{F}_{E,M} = \overline{\mathbb{F}}_{E,M} \setminus \{-\infty, +\infty, \mathrm{NaN}\}.$$

Thus $\mathbb{F}_{E,M} \subseteq \mathbb{R}$ consists of finite values.

Remark. The bit strings for $+0$ and -0 both map to the real value 0, whereas several distinct bit strings represent NaN. Thus $\gamma_{E,M}$ identifies bit strings that have the same value in this abstract model. IEEE binary32 and binary64 have $(E, M) = (8, 23)$ and $(11, 52)$, respectively; signed-zero behavior in actual IEEE arithmetic is not modeled by this value map.

Definition 3.2. Rounding
The largest finite floating-point value and the positive overflow threshold are, respectively,

$$\max \mathbb{F}_{E,M} = (2 - 2^{-M}) 2^{e_{\max}}, \qquad K_{E,M} = (2 - 2^{-M-1}) 2^{e_{\max}}.$$

We also write $F_{\max} = \max \mathbb{F}_{E,M}$ when useful. Fix *round to nearest, ties to even* throughout the notes. The rounding map $\mathrm{rnd}_{E,M} \colon \mathbb{R} \to \overline{\mathbb{F}}_{E,M}$ is

$$\mathrm{rnd}_{E,M}(x) = \begin{cases} -\infty, & x \leq -K_{E,M}, \\ \operatorname*{arg\,min}_{y \in \mathbb{F}_{E,M}} |x - y|, & |x| < K_{E,M}, \\ +\infty, & x \geq K_{E,M}. \end{cases}$$

At equal distances choose the value with an even least significant stored significand bit. In particular, a tie near zero is rounded to zero. The selected result is unique.

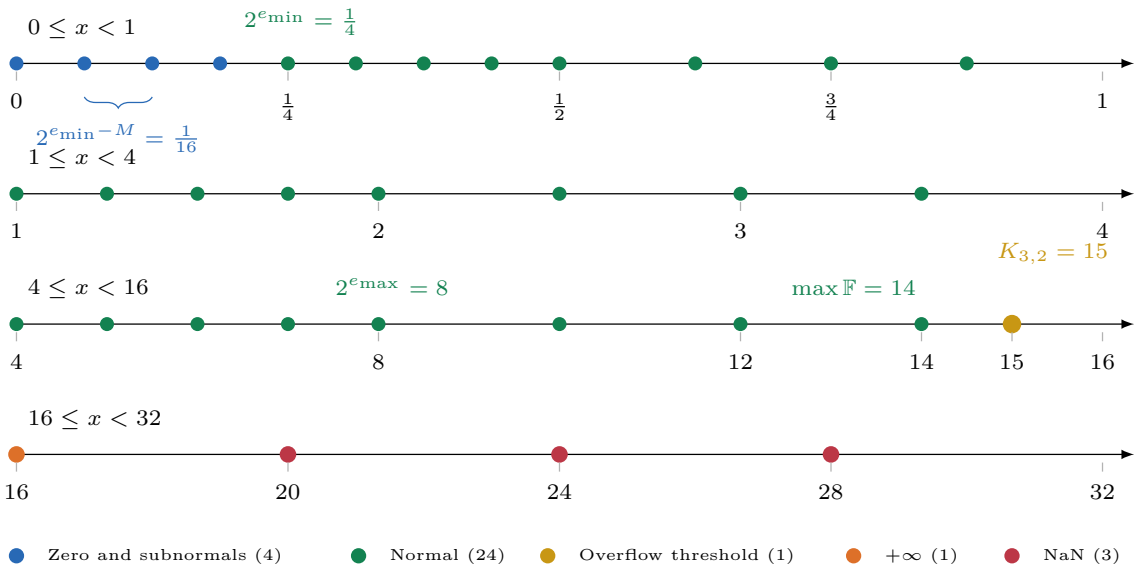


Figure 1: The nonnegative $(E, M) = (3, 2)$ format. Of the 32 nonnegative bit-pattern encodings, 24 are normal, four represent zero or a subnormal value, one represents $+\infty$, and three represent NaN. The additional yellow marker is the nonrepresentable overflow threshold $K_{3,2} = 15$; $\max \mathbb{F}_{3,2} = 14$. The final strip places special encodings in illustrative slots, not at real-number values.

Proposition 3.3. Monotonicity of Rounding
For $x, y \in \mathbb{R}$,

$$x \le y \implies \mathrm{rnd}_{E,M}(x) \le \mathrm{rnd}_{E,M}(y)$$

with the usual order on $\mathbb{R} \cup \{-\infty, +\infty\}$.

Strict monotonicity fails: distinct real values can round to the same floating-point value.

Definition 3.4. Floating-Point Operations
For $a, b, c \in \mathbb{F}_{E,M}$, define

$$\begin{aligned} a \oplus b &= \mathrm{rnd}_{E,M}(a + b), \quad & a \ominus b &= \mathrm{rnd}_{E,M}(a - b), \\ a \otimes b &= \mathrm{rnd}_{E,M}(ab), & a \oslash b &= \mathrm{rnd}_{E,M}(a/b) \quad (b \neq 0). \end{aligned}$$

Secondary notations are $\odot$ for multiplication and $\oslash$ for division. Fused multiply-add rounds only once:

$$\mathrm{fma}(a, b, c) = \mathrm{rnd}_{E,M}(ab + c).$$

These operations take values in $\overline{\mathbb{F}}_{E,M}$; operations on infinite values, NaN, or signed-zero bit patterns are outside the present finite-input model.

Remark. Algebraic Properties. On finite inputs, rounded addition and multiplication are commutative and have respective identities 0 and 1 whenever the operations remain finite. Neither operation is generally associative, and multiplication need not distribute over addition. For $(E, M) = (3, 2)$, ties-to-even gives

$$\begin{aligned} (1 \oplus \tfrac{1}{8}) \oplus (-1) = 0 \neq \tfrac{1}{8} = 1 \oplus (\tfrac{1}{8} \oplus (-1)), \\ 2 \otimes (\tfrac{3}{4} \otimes \tfrac{1}{8}) = \tfrac{1}{4} \neq \tfrac{3}{16} = (2 \otimes \tfrac{3}{4}) \otimes \tfrac{1}{8}, \\ 3 \otimes (1 \oplus \tfrac{3}{2}) = 8 \neq 7 = (3 \otimes 1) \oplus (3 \otimes \tfrac{3}{2}). \end{aligned}$$

Every $a \in \mathbb{F}_{E,M}$ has the unique additive inverse $-a$: since all floats lie on the lattice of spacing $2^{e_{\min}-M}$, $a \oplus b = 0$ holds exactly when $b = -a$. Multiplicative inverses need neither exist nor be unique; for the same $(3, 2)$ format,

$$\tfrac{3}{2} \otimes \tfrac{5}{8} = 1 = \tfrac{3}{2} \otimes \tfrac{3}{4}, \qquad \tfrac{1}{16} \otimes b \neq 1 \quad (b \in \mathbb{F}_{3,2}).$$

Finally, rounded operations may overflow to $\pm\infty$, which are outside $\mathbb{F}_{E,M}$. Hence this is not a field.

4 Floating-Point Errors

Fix a format (E, M) as in Section 3. Write $\mathbb{F} = \mathbb{F}_{E,M}$ and $\mathrm{rnd} = \mathrm{rnd}_{E,M}$, and retain $e_{\min}$, $e_{\max}$, $K_{E,M}$, and $\max \mathbb{F}$ from that section. The rounding mode remains round to nearest, ties to even.

Definition 4.1. Normal and Subnormal Ranges
A real number x is in the *normal range* if

$$2^{e_{\min}} \leq |x| \leq \max \mathbb{F},$$

and in the *subnormal range* if

$$|x| < 2^{e_{\min}}.$$

The subnormal range includes 0; a *subnormal number* is a nonzero representable value within that range.

Definition 4.2. Absolute and Relative Errors
For $x \in \mathbb{R}$ and an approximation $\widehat{x} \in \mathbb{F}$,

$$\mathrm{err}_{\mathrm{abs}}(x, \widehat{x}) \coloneqq |x - \widehat{x}|,$$
$$\mathrm{err}_{\mathrm{rel}}(x, \widehat{x}) \coloneqq \frac{|x - \widehat{x}|}{|x|} \qquad (x \neq 0).$$

Definition 4.3. Unit in the Last Place
The *unit in the last place* is $\mathrm{ulp} \colon \mathbb{R} \to \mathbb{R}_{>0}$, where

$$\mathrm{ulp}(x) \coloneqq \begin{cases} 2^{e-M}, & 2^e \leq |x| < 2^{e+1}, \quad e \in \mathbb{Z},\ e \geq e_{\min}, \\ 2^{e_{\min}-M}, & |x| < 2^{e_{\min}}. \end{cases}$$

For an approximation $\widehat{x} \in \mathbb{F}$, set

$$\mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) \coloneqq \frac{|x - \widehat{x}|}{\mathrm{ulp}(x)}.$$

It measures error in units of the local floating-point spacing. The *unit roundoff* (denoted here by machine epsilon) is

$$\varepsilon \coloneqq \tfrac{1}{2}\,\mathrm{ulp}(1) = 2^{-M-1}.$$

Proposition 4.4. Relative and ULP Errors
For $x \neq 0$ and $\widehat{x} \in \mathbb{F}$,

$$\mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) \leq \frac{\mathrm{err}_{\mathrm{rel}}(x, \widehat{x})}{\varepsilon}.$$

If x is in the normal range, also

$$\mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) \geq \frac{\mathrm{err}_{\mathrm{rel}}(x, \widehat{x})}{2\varepsilon}.$$

Proof. By definition,

$$\mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) = \mathrm{err}_{\mathrm{rel}}(x, \widehat{x}) \frac{|x|}{\mathrm{ulp}(x)}.$$

For $2^e \le |x| < 2^{e+1}$, $e \ge e_{\min}$,

$$2^M \le \frac{|x|}{\mathrm{ulp}(x)} < 2^{M+1} = \frac{1}{\varepsilon}.$$

In the subnormal range, the ratio is less than 2^M. These bounds prove both assertions. The normal-range assumption is essential: for $x = 2^{e_{\min}-M}$ and $\widehat{x} = 0$, both errors equal 1, but $1/(2\varepsilon) = 2^M > 1$. □

Proposition 4.5. Rounding Error Bounds
For $x \in \mathbb{R}$ with $|x| \le \max \mathbb{F}$:

(i) If x is in the normal range and $x \neq 0$, then

$$\mathrm{err}_{\mathrm{rel}}(x, \mathrm{rnd}(x)) \le \varepsilon.$$

(ii) If x is in the subnormal range, then

$$\mathrm{err}_{\mathrm{abs}}(x, \mathrm{rnd}(x)) \le 2^{e_{\min}} \varepsilon.$$

(iii) In either range,

$$\mathrm{err}_{\mathrm{ulp}}(x, \mathrm{rnd}(x)) \le \tfrac{1}{2}.$$

Proof. For $2^e \le |x| < 2^{e+1}$ in the normal range, the local spacing is 2^{e-M}. Nearest rounding gives

$$\frac{|x - \mathrm{rnd}(x)|}{|x|} \le \frac{2^{e-M-1}}{2^e} = \varepsilon.$$

In the subnormal range the spacing is $2^{e_{\min}-M}$, so $|x - \mathrm{rnd}(x)| \le 2^{e_{\min}-M-1}$. Both bounds come from rounding by at most half of $\mathrm{ulp}(x)$. □

Theorem 4.6. Error of Floating-Point Operations
Let $a, b \in \mathbb{F}$, with $b \neq 0$ for division, and suppose the exact result $a * b$ satisfies $|a * b| \le \max \mathbb{F}$. Pair $* \in \{+, -, \times, \div\}$ with its rounded counterpart $\circledast \in \{\oplus, \ominus, \otimes, \oslash\}$. Then the following cases hold:

(i) Normal range *($* \in \{+, -, \times, \div\}$):*

$$a \circledast b = (a * b)(1 + \delta), \qquad |\delta| \le \varepsilon.$$

(ii) Subnormal range, *$* \in \{+, -\}$:*

$$a \circledast b = a * b.$$

(iii) Subnormal range, *$* \in \{\times, \div\}$:*

$$a \circledast b = a * b + \delta', \qquad |\delta'| \le 2^{e_{\min}} \varepsilon.$$

In particular, all cases admit $a \circledast b = (a * b)(1 + \delta) + \delta'$ with the displayed bounds, where $\delta' = 0$ in (i), $\delta = \delta' = 0$ in (ii), and $\delta = 0$ in (iii).

Proof. By definition, $a \circledast b = \mathrm{rnd}(a * b)$; the normal and subnormal rounding bounds establish (i) and (iii). For (ii), every element of $\mathbb{F}$ is an integer multiple of $2^{e_{\min}-M}$. Thus $a \pm b$ also lies on this lattice, and a sum or difference in the subnormal range is exactly representable. □

Definition 4.7. Underflow and Overflow
For a nonzero exact arithmetic result $a * b$, distinguish:

(i) *Underflow range*: $0 < |a * b| < 2^{e_{\min}}$.
(ii) *Beyond the finite range*: $|a * b| > \max \mathbb{F}$.
(iii) *Overflow to infinity*: $|a * b| \geq K_{E,M}$, under the fixed rounding mode.

These are not interchangeable: an underflow-range result need not round to zero, and a value strictly between $\max \mathbb{F}$ and $K_{E,M}$ may still round to $\max \mathbb{F}$ (with the appropriate sign).

Example 4.8. Error in Repeated Multiplication
Let $a, b, c \in \mathbb{F} \setminus \{0\}$, and suppose that ab, $(a \otimes b)c$, and abc are in the normal range, with no intermediate overflow. Then

$$(a \otimes b) \otimes c = abc(1 + \delta_1)(1 + \delta_2), \qquad |\delta_1|, |\delta_2| \leq \varepsilon.$$

Consequently,

$$\begin{aligned} \mathrm{err}_{\mathrm{rel}}(abc, (a \otimes b) \otimes c) &\leq 2\varepsilon + \varepsilon^2, \\ \mathrm{err}_{\mathrm{ulp}}(abc, (a \otimes b) \otimes c) &\leq 2 + \varepsilon. \end{aligned}$$

The leading-order relative error is at most 2ε.

Example 4.9. Error in Repeated Addition
Let $a, b, c \in \mathbb{F}$ with $a + b + c \neq 0$ and suppose $|a + b|, |(a \oplus b) + c| \leq \max \mathbb{F}$. Rounded addition has no additive subnormal error, so

$$\begin{aligned} (a \oplus b) \oplus c &= \big((a + b)(1 + \delta_1) + c\big)(1 + \delta_2) \\ &= (a + b + c)(1 + \delta_2) + (a + b)\delta_1(1 + \delta_2), \end{aligned}$$

where $|\delta_1|, |\delta_2| \leq \varepsilon$. Therefore

$$\begin{aligned} &\mathrm{err}_{\mathrm{rel}}(a + b + c, (a \oplus b) \oplus c) \\ &\qquad \leq \varepsilon \left(1 + \frac{|a + b|}{|a + b + c|}\right) + \varepsilon^2 \frac{|a + b|}{|a + b + c|}, \\ &\mathrm{err}_{\mathrm{ulp}}(a + b + c, (a \oplus b) \oplus c) \\ &\qquad \leq 1 + \frac{|a + b|}{|a + b + c|} + \varepsilon \frac{|a + b|}{|a + b + c|}. \end{aligned}$$

The ratio $|a + b|/|a + b + c|$ can be large. For instance, when $2^{M+1} \in \mathbb{F}$, put $a = 1$, $b = 2^{M+1}$, and $c = -2^{M+1}$. Ties-to-even gives

$$(a \oplus b) \oplus c = 0, \qquad a + b + c = 1,$$

so the relative error is 1 and the ULP error is 2^M.

Remark. Catastrophic Cancellation. When large, nearly equal quantities cancel, small errors inherited from intermediate calculations may dominate their difference. The preceding example loses the small term a in its first addition. The effect can also expose pre-existing input errors: if $\widehat{x}, \widehat{y} \in \mathbb{F}$ approximate $x, y \in \mathbb{R}$, then even when subtracting the represented operands is exact,

$$|(\widehat{x} - \widehat{y}) - (x - y)| \leq |\widehat{x} - x| + |\widehat{y} - y|.$$

Dividing by $|x - y|$ or $\mathrm{ulp}(x - y)$ illustrates how relative or ULP error can be large when $x \approx y$, independently of whether the final subtraction incurs rounding.

Example 4.10. Stable Evaluation of $a^2 - b^2$
For $a, b \in \mathbb{F}$ with $a^2 \neq b^2$, compare

$$(a \otimes a) \ominus (b \otimes b), \qquad (a \ominus b) \otimes (a \oplus b).$$

Assume all exact intermediate results are normal and within the finite range. For some $|\delta_i| \leq \varepsilon$, the direct calculation gives

$$(a \otimes a) \ominus (b \otimes b) = (a^2(1+\delta_1) - b^2(1+\delta_2))(1+\delta_3).$$

Its relative error is therefore bounded by

$$\left(1 + \frac{a^2 + b^2}{|a^2 - b^2|}\right)\varepsilon + O(\varepsilon^2).$$

By contrast, the factored calculation satisfies

$$(a \ominus b) \otimes (a \oplus b) = (a-b)(a+b)(1+\delta_1)(1+\delta_2)(1+\delta_3),$$

with relative error at most $3\varepsilon + O(\varepsilon^2)$. For a normal exact result, the corresponding ULP-error bounds are respectively $1 + (a^2 + b^2)/|a^2 - b^2| + O(\varepsilon)$ and $3 + O(\varepsilon)$. The former can be large when $|a| \approx |b|$; factoring avoids subtracting two rounded squares.

Remark. An ULP Bound. Suppose $\mathrm{err}_{\mathrm{ulp}}(r, r') \leq 1$ and $\mathrm{rnd}(r')$ is finite. Then

$$\mathrm{err}_{\mathrm{ulp}}(r, \mathrm{rnd}(r')) \leq \mathrm{err}_{\mathrm{ulp}}(r, r') + \tfrac{1}{2}.$$

If $\mathrm{ulp}(r') \leq \mathrm{ulp}(r)$, apply the triangle inequality and the half-ULP rounding bound. Otherwise r' lies just across a power-of-two boundary from r. The one-ULP assumption keeps r' in the first half of the new binade, so it rounds to that boundary, no farther from r than r'.

Example 4.11. Stable Quadratic Roots
Let $a \neq 0$ and $\Delta = b^2 - 4ac \geq 0$. The quadratic roots are

$$x_{\pm} = \frac{-b \pm \sqrt{\Delta}}{2a}.$$

When $b^2 \gg |4ac|$, the numerator $-b - \sqrt{\Delta}$ for $b < 0$ (or $-b + \sqrt{\Delta}$ for $b > 0$) subtracts nearly equal quantities and may suffer cancellation. Rationalization gives the exact alternatives

$$\frac{-b - \sqrt{\Delta}}{2a} = \frac{2c}{-b + \sqrt{\Delta}} \quad (b < 0),$$
$$\frac{-b + \sqrt{\Delta}}{2a} = \frac{2c}{-b - \sqrt{\Delta}} \quad (b > 0),$$

provided the displayed denominators are nonzero.

An independent problem is that forming b^2 may overflow even when both roots are finite. For $b \neq 0$ the exact identity

$$\sqrt{b^2 - 4ac} = |b|\sqrt{1 - \frac{4ac}{b^2}}$$

suggests rescaling, although a real implementation must still protect the intermediate quotient. When $|4ac/b^2| \ll 1$,

$$\sqrt{1-t} = 1 - \tfrac{1}{2}t - \tfrac{1}{8}t^2 + O(t^3)$$

explains the cancellation and yields, for example,

$$x_- \approx \begin{cases} -b/a + c/b, & b > 0, \\ -c/b, & b < 0. \end{cases}$$

These are asymptotic approximations, not exact identities.

Example 4.12. Two Ways to Compute a Mean
For $a, b \in \mathbb{F}$, compare

$$(a \oplus b) \oslash 2, \qquad (a \oslash 2) \oplus (b \oslash 2).$$

If $|a + b| \le \max \mathbb{F}$, the first is correctly rounded:

$$(a \oplus b) \oslash 2 = \mathrm{rnd}\left(\frac{a+b}{2}\right), \qquad \min(a, b) \le (a \oplus b) \oslash 2 \le \max(a, b).$$

But computing $a \oplus b$ may overflow even when the exact mean is representable. The second avoids this intermediate overflow. In particular, if $a/2, b/2 \in \mathbb{F}$, both halves are exact and the second expression is correctly rounded. When halving incurs underflow rounding, its ULP error is at most 1, rather than necessarily zero.

Theorem 4.13. Correctly Rounded Mean
Let

$$C \in [2^{e_{\min}+M+3}, 2^{e_{\max}}] \cap \mathbb{F}.$$

For $a, b \in \mathbb{F}$, define

$$m(a, b) = \begin{cases} (a \oplus b) \oslash 2, & |a| < C \text{ and } |b| < C, \\ (a \oslash 2) \oplus (b \oslash 2), & \text{otherwise.} \end{cases}$$

Then, whenever the displayed floating-point operations are defined,

$$m(a, b) = \mathrm{rnd}\left(\frac{a+b}{2}\right) \qquad (a, b \in \mathbb{F}).$$

5 Error-Free Computations

Fix $\mathbb{F} = \mathbb{F}_{E,M}$ and $\mathrm{rnd} = \mathrm{rnd}_{E,M}$. For $a \in \mathbb{F} \setminus \{0\}$, let $e_a \in [e_{\min}, e_{\max}] \cap \mathbb{Z}$ denote its floating-point exponent, with $e_a = e_{\min}$ for subnormals; put $e_0 = e_{\min}$ when needed. Set $\Delta_{\min} := 2^{e_{\min}-M}$. All exactness assertions concern finite results under their stated range conditions.

Lemma 5.1. Integer Significands
For $a \in \mathbb{R}$,

$$a \in \mathbb{F} \iff \exists m, e \in \mathbb{Z} : \quad a = m2^{e-M}, \quad |m| < 2^{M+1}, \quad e_{\min} \le e \le e_{\max}.$$

In particular, every $a \in \mathbb{F}$ is an integer multiple of $\Delta_{\min}$.

Proof. For a normal value, take $e = e_a$ and let m be its signed $(M + 1)$-bit significand. For a subnormal, take $e = e_{\min}$ and $|m| < 2^M$; zero is immediate. Conversely, normalize the integer m by shifting powers of two into the exponent until either the leading bit is in the normal significand position or the exponent reaches $e_{\min}$. Since $|m| < 2^{M+1}$, no more than $M + 1$ significant bits are required, and the exponent is not raised beyond $e_{\max}$. The resulting value is normal, subnormal, or zero. □

Theorem 5.2. Hauser: Exact Addition
For $a, b \in \mathbb{F}$,

$$|a + b| \le 2^{e_{\min}+1} \implies a \oplus b = a + b.$$

Proof. Both inputs lie in $\Delta_{\min}\mathbb{Z}$; hence so does $a + b$. Every multiple of $\Delta_{\min}$ of absolute value at most $2^{e_{\min}+1}$ is representable, so rounding is exact. □

Theorem 5.3. Sterbenz: Exact Subtraction
For $a, b \in \mathbb{F}$,

$$\frac{a}{2} \le b \le 2a \implies a \ominus b = a - b.$$

Equivalently, one may require $b/2 \le a \le 2b$.

Proof. The inequalities imply $a, b \ge 0$. The case $ab = 0$ is immediate. By symmetry assume $a \ge b$; then $b \le a \le 2b$. By the integer-significand lemma,

$$a - b = (m_a 2^{e_a - e_b} - m_b) 2^{e_b - M} = m' 2^{e_b - M}, \qquad m' \in \mathbb{Z}.$$

Here $e_a \ge e_b$, and $0 \le a - b \le b$ implies $0 \le m' \le m_b < 2^{M+1}$. The lemma yields $a - b \in \mathbb{F}$; subtraction is therefore exact. □

Thus Sterbenz guarantees exact subtraction of nearby *represented* floats, but does not prevent amplification of errors already present in those operands (Section 4).

Proposition 5.4. Exact Scaling by Powers of Two
Suppose $a, b \in \mathbb{F} \setminus \{0\}$ satisfy $|a| = (\alpha_0.\alpha_1 \dots \alpha_M)_2 \, 2^{e_a}$ and $|b| = 2^{e_b}$. If $e_a + e_b \in [e_{\min}, e_{\max}]$, then $a \otimes b = ab$. Similarly, if $e_a - e_b \in [e_{\min}, e_{\max}]$, then $a \oslash b = a/b$.

Proposition 5.5. Exact Products with Trailing Zeros
Let $a, b \in \mathbb{F}$ be nonzero normal values whose binary significands have at least j and j' trailing zero bits. If

$$j + j' \ge M + 1, \qquad e_{\min} \le e_a + e_b \le e_{\max} - 1,$$

then $ab \in \mathbb{F}$ and $a \otimes b = ab$.

Proof. After removing the trailing zeros, the integer significands have at most $M + 1 - j$ and $M + 1 - j'$ bits. Their product therefore has at most

$$2(M + 1) - j - j' \le M + 1$$

significant bits. Its normalized exponent is $e_a + e_b$ or $e_a + e_b + 1$, both within the normal exponent range. Hence the exact product is representable. □

Theorem 5.6. Dekker's Theorem (Fast2Sum)
For $a, b \in \mathbb{F}$ with $|a + b| \le \max \mathbb{F}$, put

$$s = a \oplus b, \qquad r = (a + b) - s.$$

Then $r \in \mathbb{F}$. If $e_a \ge e_b$, the algorithm

$$\begin{aligned} s &= a \oplus b, \\ b' &= s \ominus a, \qquad \text{(fast2sum)} \\ z &= b \ominus b' \end{aligned}$$

returns $(s, z) = (a \oplus b, r)$; in particular, $a + b = s + z$ exactly.

Proof. We prove two representability claims:

$$s - a \in \mathbb{F}, \qquad b - (s - a) \in \mathbb{F}.$$

By symmetry in a and b, we may assume $e_a \ge e_b$ when proving the representability of r. The result is immediate if $a = 0$ or $b = 0$. Since rounding is odd, assume $a > 0$.

First, $s - a \in \mathbb{F}$. If $b < 0$ and $e_a = e_b = e_{\min}$, Hauser makes the sum exact. If $b < 0$ and $e_a = e_b > e_{\min}$, the two magnitudes differ by a factor below two, so Sterbenz again makes the sum exact. If $b < 0$ and $e_a > e_b$, then $a > |b|$. For $|b| \ge a/2$, Sterbenz gives an exact

sum; otherwise $a/2 \in \mathbb{F}$ and monotonicity yields $a/2 \le s \le a$, so Sterbenz makes $s - a$ exactly representable.

If $b > 0$ and $e_a > e_b$, then $b < a$ and $a \le s \le 2a$; Sterbenz implies $s - a \in \mathbb{F}$. Finally, if $b > 0$ and $e_a = e_b$, put $h = 2^{e_a - M}$. Both inputs and s are multiples of h. The sum rounds with error at most h, even if it crosses into the next binade. Consequently,

$$0 \le s - a \le b + h \le 2^{e_a+1}.$$

The integer-significand lemma gives $s - a \in \mathbb{F}$ (the endpoint is itself representable). Hence in all cases,

$$b' = s \ominus a = s - a.$$

Second, $r = b - (s - a) \in \mathbb{F}$. If the sum is exact, $r = 0$. Otherwise set $h_b = 2^{e_b - M}$. The inputs a, b lie on $h_b\mathbb{Z}$. If the rounded sum had spacing smaller than h_b, their exact sum, itself a multiple of h_b, would already be representable. Thus, in the inexact case, s is also a multiple of h_b. It follows that $r \in h_b\mathbb{Z}$. Nearest rounding, compared with the representable candidate a, gives

$$|r| = |(a + b) - s| \le |(a + b) - a| = |b|.$$

The integer-significand lemma now makes r representable. Thus $z = b \ominus b' = r$, and $a + b = s + z$ exactly. □

Theorem 5.7. Boldo's Theorem (Product Residual)
For $a, b \in \mathbb{F}$ with $|ab| \le \max \mathbb{F}$, put $s = a \otimes b$ and $r = ab - s$. If

$$e_a + e_b \ge e_{\min} + M,$$

then $r \in \mathbb{F}$ and

$$r = \mathrm{fma}(a, b, -s).$$

Proof. The case $ab = 0$ is immediate. By the integer-significand lemma, $a = m_a 2^{e_a - M}$ and $b = m_b 2^{e_b - M}$. Therefore the exact product belongs to the binary lattice

$$ab \in h\mathbb{Z}, \qquad h = 2^{e_a + e_b - 2M} \ge \Delta_{\min}.$$

If this product is not representable, the rounding grid at its magnitude is coarser than h; otherwise the lattice point ab would be exactly representable. In the inexact case the rounded product s is also a multiple of h. Thus in either case $r = kh$ for some integer k.

Since $|a| < 2^{e_a+1}$ and $|b| < 2^{e_b+1}$, we have $|ab| < 2^{e_a+e_b+2}$. Nearest rounding bounds the residual by half the spacing in the binade of the exact product:

$$|r| \le 2^{e_a+e_b-M}, \qquad |k| \le 2^M.$$

Writing $r = k2^{(e_a+e_b-M)-M}$, the exponent $e_a + e_b - M$ is at least $e_{\min}$ by hypothesis. It is at most $e_{\max}$ under $|ab| \le \max \mathbb{F}$: for normal inputs $e_a + e_b \le e_{\max}$, and if an input is subnormal, its assigned exponent is $e_{\min} \le 0$. Hence the integer-significand lemma implies $r \in \mathbb{F}$. A fused multiply-add rounds the exact quantity $ab - s = r$ once, so $\mathrm{fma}(a, b, -s) = \mathrm{rnd}(r) = r$. □

6 Automatic Error Analysis

Definition 6.1. Floating-Point Expression Syntax
Fix $N \in \mathbb{N}$ and $\mathbb{F} = \mathbb{F}_{E,M}$. Expressions are generated by the grammar

$$e ::= c \mid x_i \mid -e \mid e + e \mid e \times e, \qquad c \in \mathbb{F}, \quad 1 \le i \le N.$$

An expression is a syntactic object, not yet a function.

Definition 6.2. Floating-Point Semantics
Let $D_e \subseteq \mathbb{F}^N$ be the set of input valuations for which every rounded operation in e has a finite result. For $\sigma = (\sigma_1, \ldots, \sigma_N) \in D_e$, the floating-point semantics $[\![e]\!] \colon D_e \to \mathbb{F}$ is given recursively by

$$\begin{aligned} [\![c]\!](\sigma) &= c, & [\![x_i]\!](\sigma) &= \sigma_i, \\ [\![-e]\!](\sigma) &= -[\![e]\!](\sigma), & [\![e_1 + e_2]\!](\sigma) &= [\![e_1]\!](\sigma) \oplus [\![e_2]\!](\sigma), \\ [\![e_1 \times e_2]\!](\sigma) &= [\![e_1]\!](\sigma) \otimes [\![e_2]\!](\sigma). \end{aligned}$$

For constants and variables, $D_e = \mathbb{F}^N$; the other constructors restrict the domain to valuations for which their subexpressions and the indicated rounded operation remain finite.

Definition 6.3. Real-Valued Semantics
The function $[\![e]\!]_{\mathrm{r}} \colon \mathbb{F}^N \to \mathbb{R}$ is defined by the same recursion, replacing $\oplus$ and $\otimes$ by exact $+$ and $\times$. Both interpretations use the same input valuation σ.

Example 6.4. Different Semantics
For $e = (1 + x_1) + (-x_1)$, suppose $2^{M+1} \in \mathbb{F}$ and take $\sigma_1 = 2^{M+1}$. Then

$$[\![e]\!](\sigma) = 0, \qquad [\![e]\!]_{\mathrm{r}}(\sigma) = 1.$$

Definition 6.5. Sound Range and Error Analysis
Fix the admissible input set

$$\Sigma \coloneqq \prod_{i=1}^{N} ([l_i, r_i] \cap \mathbb{F}) \subseteq D_e, \qquad l_i, r_i \in \mathbb{F}, \quad l_i \le r_i.$$

The inclusion excludes intermediate overflow for inputs in Σ. Assume also that rounding of endpoints used by the analysis remains finite. A sound *range analysis* and *absolute-error analysis* assign respectively an interval $\mathrm{range}(e)$ and a bound $\mathrm{err}(e) \ge 0$ such that

$$\begin{aligned} [\![e]\!](\sigma) &\in \mathrm{range}(e), \\ |[\![e]\!](\sigma) - [\![e]\!]_{\mathrm{r}}(\sigma)| &\le \mathrm{err}(e) \end{aligned} \qquad (\forall \sigma \in \Sigma).$$

The fixed Σ is suppressed in the notation. We compute both bounds recursively from expression syntax.

Definition 6.6. Recursive Range Bounds
For constants and variables, set

$$\mathrm{range}(c) = [c, c], \qquad \mathrm{range}(x_i) = [l_i, r_i].$$

If $\mathrm{range}(e) = [a, b]$, put $\mathrm{range}(-e) = [-b, -a]$. For $\mathrm{range}(e_i) = [a_i, b_i]$, $i = 1, 2$, set

$$\mathrm{range}(e_1 + e_2) = [\mathrm{rnd}(a_1 + a_2), \mathrm{rnd}(b_1 + b_2)].$$

Indeed, monotonicity of exact addition and of rounding gives

$$a_1 + a_2 \le [\![e_1]\!](\sigma) + [\![e_2]\!](\sigma) \le b_1 + b_2,$$

which proves the recursive enclosure.

Definition 6.7. Recursive Absolute-Error Bounds
Set

$$\mathrm{err}(c) = \mathrm{err}(x_i) = 0, \qquad \mathrm{err}(-e) = \mathrm{err}(e).$$

For $\mathrm{range}(e_i) = [a_i, b_i]$, $i = 1, 2$, define

$$\begin{aligned} \mathrm{err}(e_1 + e_2) = {} & \mathrm{err}(e_1) + \mathrm{err}(e_2) \\ & + \tfrac{1}{2}\,\mathrm{ulp}(\max\{|a_1 + a_2|, |b_1 + b_2|\}). \end{aligned}$$

To verify this rule, write $u_i = [\![e_i]\!](\sigma)$ and $v_i = [\![e_i]\!]_{\mathrm{r}}(\sigma)$. The triangle inequality and half-ULP

bound give

$$\begin{aligned}|\operatorname{rnd}(u_1+u_2)-(v_1+v_2)| &\le |\operatorname{rnd}(u_1+u_2)-(u_1+u_2)| \\ &\quad + |u_1-v_1|+|u_2-v_2| \\ &\le \tfrac{1}{2}\operatorname{ulp}(u_1+u_2)+\operatorname{err}(e_1)+\operatorname{err}(e_2).\end{aligned}$$

The monotonicity of ulp in $|x|$ and the range bounds justify the displayed recursion.

Definition 6.8. Multiplication Bounds
For $\operatorname{range}(e_i)=[a_i,b_i]$, $i=1,2$, put

$$\begin{aligned}p_- &\coloneqq \min\{a_1a_2,a_1b_2,b_1a_2,b_1b_2\},\\ p_+ &\coloneqq \max\{a_1a_2,a_1b_2,b_1a_2,b_1b_2\}.\end{aligned}$$

Then

$$\operatorname{range}(e_1\times e_2)=[\operatorname{rnd}(p_-),\operatorname{rnd}(p_+)].$$

Writing $U_i=\max\{|a_i|,|b_i|\}$ and $\eta_i=\operatorname{err}(e_i)$, define

$$\begin{aligned}\operatorname{err}(e_1\times e_2) &= U_1\eta_2+U_2\eta_1+\eta_1\eta_2 \\ &\quad + \tfrac{1}{2}\operatorname{ulp}(\max\{|p_-|,|p_+|\}).\end{aligned}$$

The four endpoint products bound all exact products from the two intervals; rounding preserves that enclosure. For $u_i=[\![e_i]\!](\sigma)$ and $v_i=[\![e_i]\!]_{\mathrm{r}}(\sigma)$,

$$\begin{aligned}|\operatorname{rnd}(u_1u_2)-v_1v_2| &\le |\operatorname{rnd}(u_1u_2)-u_1u_2|+|u_1u_2-v_1v_2| \\ &\le \tfrac{1}{2}\operatorname{ulp}(\max\{|p_-|,|p_+|\}) \\ &\quad + U_1\eta_2+U_2\eta_1+\eta_1\eta_2,\end{aligned}$$

proving soundness of the error rule.

Definition 6.9. Square-Root Extension
The grammar may be extended by $\sqrt{e}$ on valuations where both semantics of e are nonnegative. Define

$$\begin{aligned}[\![\sqrt{e}]\!](\sigma) &= \operatorname{rnd}\left(\sqrt{[\![e]\!](\sigma)}\right),\\ [\![\sqrt{e}]\!]_{\mathrm{r}}(\sigma) &= \sqrt{[\![e]\!]_{\mathrm{r}}(\sigma)}.\end{aligned}$$

If $\operatorname{range}(e)=[a,b]\subseteq[0,\max\mathbb{F}]$, set

$$\begin{aligned}\operatorname{range}(\sqrt{e}) &= [\operatorname{rnd}(\sqrt{a}),\operatorname{rnd}(\sqrt{b})],\\ \operatorname{err}(\sqrt{e}) &= \sqrt{\operatorname{err}(e)}+\tfrac{1}{2}\operatorname{ulp}(\sqrt{b}).\end{aligned}$$

Range soundness follows from the monotonicity of square root and rounding. For $u,v\ge 0$,

$$|\sqrt{u}-\sqrt{v}|\le\sqrt{|u-v|}.$$

Together with the triangle inequality and half-ULP rounding, this proves the error rule.

Remark. Limitations of Interval Analysis. Independent interval bounds lose correlations between repeated occurrences of a variable. For $e=x_1-x_1$, understood as $x_1+(-x_1)$, and $x_1\in[-1,1]$, interval propagation gives $[-2,2]$, although both semantics are identically zero. Soundness does not imply sharpness: an error upper bound may be positive when the actual rounding error vanishes.